

SIM1003 Calculus II
Tutorial 9

All even-numbered questions in the following exercises

Exercise 11.3:

Dot Product and Projections

In Exercises 1–8, find (a) $\mathbf{v} \cdot \mathbf{u}$, $|\mathbf{v}|$, $|\mathbf{u}|$.

(b) the cosine of the angle between $\mathbf{v}$ and $\mathbf{u}$.

(c) the scalar component of $\mathbf{u}$ in the direction of $\mathbf{v}$.

(d) the vector $\text{proj}_{\mathbf{v}} \mathbf{u}$.

1. $\mathbf{v} = 2\mathbf{i} - 4\mathbf{j} + \sqrt{5}\mathbf{k}$, $\mathbf{u} = -2\mathbf{i} + 4\mathbf{j} - \sqrt{5}\mathbf{k}$

2. $\mathbf{v} = (3/5)\mathbf{i} + (4/5)\mathbf{k}$, $\mathbf{u} = 5\mathbf{i} + 12\mathbf{j}$

3. $\mathbf{v} = 10\mathbf{i} + 11\mathbf{j} - 2\mathbf{k}$, $\mathbf{u} = 3\mathbf{j} + 4\mathbf{k}$

4. $\mathbf{v} = 2\mathbf{i} + 10\mathbf{j} - 11\mathbf{k}$, $\mathbf{u} = 2\mathbf{i} + 2\mathbf{j} + \mathbf{k}$

5. $\mathbf{v} = 5\mathbf{j} - 3\mathbf{k}$, $\mathbf{u} = \mathbf{i} + \mathbf{j} + \mathbf{k}$

6. $\mathbf{v} = -\mathbf{i} + \mathbf{j}$, $\mathbf{u} = \sqrt{2}\mathbf{i} + \sqrt{3}\mathbf{j} + 2\mathbf{k}$

7. $\mathbf{v} = 5\mathbf{i} + \mathbf{j}$, $\mathbf{u} = 2\mathbf{i} + \sqrt{17}\mathbf{j}$

8. $\mathbf{v} = \langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{3}} \rangle$, $\mathbf{u} = \langle \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{3}} \rangle$

Angle Between Vectors

Find the angles between the vectors in Exercises 9–12 to the nearest hundredth of a radian.

9. $\mathbf{u} = 2\mathbf{i} + \mathbf{j}$, $\mathbf{v} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$

10. $\mathbf{u} = 2\mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\mathbf{v} = 3\mathbf{i} + 4\mathbf{k}$

11. $\mathbf{u} = 3\mathbf{i} - 7\mathbf{j}$, $\mathbf{v} = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k}$

12. $\mathbf{u} = \mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$, $\mathbf{v} = -\mathbf{i} + \mathbf{j} + \mathbf{k}$

27. (a) **Cauchy-Schwarz inequality**

Since $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}| |\mathbf{v}| \cos \theta$, show that the inequality $|\mathbf{u} \cdot \mathbf{v}| \leq |\mathbf{u}| |\mathbf{v}|$ holds for any vectors $\mathbf{u}$ and $\mathbf{v}$.

(b) Under what circumstances, if any, does $|\mathbf{u} \cdot \mathbf{v}|$ equal $|\mathbf{u}| |\mathbf{v}|$? Give reasons for your answer.

29. **Orthogonal unit vectors** If $\mathbf{u}_1$ and $\mathbf{u}_2$ are orthogonal unit vectors and $\mathbf{v} = a\mathbf{u}_1 + b\mathbf{u}_2$, find $\mathbf{v} \cdot \mathbf{u}_1$.

Exercise 11.4:

Triangles in Space

In Exercises 15–18, find

(a) Find the area of the triangle determined by the points P , Q , and R .

(b) Find a unit vector perpendicular to plane PQR .

15. $P(1, -1, 2)$, $Q(2, 0, -1)$, $R(0, 2, 1)$

16. $P(1, 1, 1)$, $Q(2, 1, 3)$, $R(3, -1, 1)$

17. $P(2, -2, 1)$, $Q(3, -1, 2)$, $R(3, -1, 1)$

18. $P(-2, 2, 0)$, $Q(0, 1, -1)$, $R(-1, 2, -2)$

Triple Scalar Products

In Exercises 19–22, find the volume of the parallelepiped (box) determined by $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$.

19. $\mathbf{u} = 2\mathbf{i}$, $\mathbf{v} = 2\mathbf{j}$, $\mathbf{w} = 2\mathbf{k}$

20. $\mathbf{u} = \mathbf{i} - \mathbf{j} + \mathbf{k}$, $\mathbf{v} = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$, $\mathbf{w} = -\mathbf{i} + 2\mathbf{j} - \mathbf{k}$

21. $\mathbf{u} = 2\mathbf{i} + \mathbf{j}$, $\mathbf{v} = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$, $\mathbf{w} = \mathbf{i} + 2\mathbf{k}$

22. $\mathbf{u} = \mathbf{i} + \mathbf{j} - 2\mathbf{k}$, $\mathbf{v} = -\mathbf{i} - \mathbf{k}$, $\mathbf{w} = 2\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$

Area of a Parallelogram

Find the areas of the parallelograms whose vertices are given in Exercises 39–40.

39. $A(0, 0, 0)$, $B(3, 2, 4)$, $C(5, 1, 4)$, $D(2, -1, 0)$

40. $A(1, 0, -1)$, $B(1, 7, 2)$, $C(2, 4, -1)$, $D(0, 3, 2)$

Area of a Triangle

Find the areas of the triangles whose vertices are given in Exercises 45–48.

45. $A(1, 0, 0)$, $B(0, 2, 0)$, $C(0, 0, -1)$

46. $A(0, 0, 0)$, $B(-1, 1, -1)$, $C(3, 0, 3)$

47. $A(1, -1, 1)$, $B(0, 1, 1)$, $C(1, 0, -1)$

48. Find the volume of a parallelepiped with one of its eight vertices at $A(0, 0, 0)$ and three adjacent vertices at $B(1, 2, 0)$, $C(0, -3, 2)$, and $D(3, -4, 5)$.

Exercise 11.5:

Lines and Line Segments

Find parametric equations for the lines in Exercises 1–12.

1. The line through the point $P(3, -4, -1)$ parallel to the vector $\mathbf{i} + \mathbf{j} + \mathbf{k}$.
2. The line through $P(1, 2, -1)$ and $Q(-1, 0, 1)$.
3. The line through $P(-2, 0, 3)$ and $Q(3, 5, -2)$.
4. The line through $P(1, 2, 0)$ and $Q(1, 1, -1)$.
5. The line through the origin parallel to the vector $2\mathbf{j} + \mathbf{k}$.
6. The line through the point $(3, -2, 1)$ parallel to the line $x = 1 + 2t, y = 2 - t, z = 3t$.
7. The line through $(1, 1, 1)$ parallel to the z -axis.
8. The line through $(2, 4, 5)$ perpendicular to the plane $3x + 7y - 5z = 21$.
9. The line through $(0, -7, 0)$ perpendicular to the plane $x + 2y + 2z = 13$.
10. The line through $(2, 3, 0)$ perpendicular to the vectors $\mathbf{u} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ and $\mathbf{v} = 3\mathbf{i} + 4\mathbf{j} + 5\mathbf{k}$.
11. The x -axis.
12. The z -axis.

Planes

Find equations for the planes in Exercises 21–26.

21. The plane through $P_0(0, 2, -1)$ normal to $\mathbf{n} = 3\mathbf{i} - 2\mathbf{j} - \mathbf{k}$.
22. The plane through $(1, -1, 3)$ parallel to the plane $3x + y + z = 7$.
23. The plane through $(1, 1, -1)$, $(2, 0, 2)$, and $(0, -2, 1)$.
24. The plane through $(2, 4, 5)$, $(1, 5, 7)$, and $(-1, 6, 8)$.
25. The plane through $P_0(2, 4, 5)$ perpendicular to the line $x = 5 + t, y = 1 + 3t, z = 4t$.
26. The plane through $A(1, -2, 1)$ perpendicular to the vector from the origin to A .
27. Find the point of intersection of the lines
 $x = 2t + 1, y = 3t + 2, z = 4t + 3$, and $x = s + 2, y = 2s + 4, z = -4s - 1$, and t
and then find the plane determined by these lines.
28. Find the point of intersection of the lines
 $x = t, y = -t + 2, z = t + 1$, and $x = 2s + 2, y = s + 3, z = 5s + 6$,
and then find the plane determined by these lines.

In Exercises 29 and 30, find the plane containing the intersecting lines.

29. $L_1: x = -1 + t, y = 2 + t, z = 1 - t; t \in \mathbb{R}.$

$L_2: x = 1 - 4s, y = 1 + 2s, z = 2 - 2s; s \in \mathbb{R}.$

30. $L_1: x = t, y = 3 - 3t, z = -2 - t; t \in \mathbb{R}.$

$L_2: x = 1 + s, y = 4 + s, z = -1 + s; s \in \mathbb{R}.$

31. Find a plane through $P_0(2, 1, -1)$ and perpendicular to the line of intersection of the planes $2x + y - z = 3$ and $x + 2y + z = 2$.

32. Find a plane through the points $P_1(1, 2, 3)$ and $P_2(3, 2, 1)$ and perpendicular to the plane $4x - y + 2z = 7$.

Distances

In Exercises 35–38, find the distance from the point to the line.

35. $(2, 1, 3); x = 2 + 2t, y = 1 + 6t, z = 3.$

36. $(2, 1, -1); x = 2t, y = 1 + 2t, z = 2t.$

37. $(3, -1, 4); x = 4 - t, y = 3 + 2t, z = -5 + 3t.$

38. $(-1, 4, 3); x = 10 + 4t, y = -3, z = 4t.$

In Exercises 41–44, find the distance from the point to the plane.

41. $(0, 1, 1); 4y + 3z = -12.$

42. $(2, 2, 3); 2x + y + 2z = 4.$

43. $(0, -1, 0); 2x + y + 2z = 4.$

44. $(1, 0, -1); -4x + y + z = 4.$

45. Find the distance from the plane $x + 2y + 6z = 1$ to the plane $x + 2y + 6z = 10$.

46. Find the distance from the line $x = 2 + t, y = 1 + t, z = -1/2 - t/2$ to the plane $x + 2y + 6z = 10$.

Angles

In Exercises 47 and 48, find the angles between the planes.

47. $x + y = 1, 2x + y - 2z = 2$

48. $5x + y - z = 10, x - 2y + 3z = -1$

In Exercises 49 and 50, find the acute angles between the intersecting lines.

49. $x = t, y = 2t, z = -t$ and $x = 1 - t, y = 5 + t, z = 2t$

50. $x = 2 + t, y = 4t + 2, z = 1 + t$ and $x = 3t - 2, y = -2, z = 2 - 2t$

In Exercises 51 and 52, find the acute angles between the lines and planes.

51. $x = 1 - t, y = 3t, z = 1 + t; 2x - y + 3z = 6$

52. $x = 2, y = 3 + 2t, z = 1 - 2t; x - y + z = 0$

In Exercises 57–60, find the point in which the line meets the plane.

57. $x = 1 - t$, $y = 3t$, $z = 1 + t$; $2x - y + 3z = 6$

58. $x = 2$, $y = 3 + 2t$, $z = -2 - 2t$; $6x + 3y - 4z = -12$

59. $x = 1 + 2t$, $y = 1 + 5t$, $z = 3t$; $x + y + z = 2$

60. $x = -1 + 3t$, $y = -2$, $z = 5t$; $2x - 3z = 7$

Find parametrizations for the lines in which the planes in Exercises 61–66 intersect.

61. $x + y + z = 1$ and $x + y = 2$

62. $3x - 6y - 2z = 3$ and $2x + y - 2z = 2$

63. $x - 2y + 4z = 2$ and $x + y - 2z = 5$

64. $5x - 2y = 11$ and $4y - 5z = -17$

Given two lines in space, either they are parallel, they intersect, or they are skew (lie in parallel planes). In Exercises 65 and 66, determine whether the lines, taken two at a time, are parallel, intersect, or are skew. If they intersect, find the point of intersection. Otherwise, find the distance between the two lines.

65. L_1 : $x = 3 + 2t$, $y = -1 + 4t$, $z = 2 - t$; $t \in \mathbb{R}$.

L_2 : $x = 1 + 4s$, $y = 1 + 2s$, $z = -3 + 4s$; $s \in \mathbb{R}$.

L_3 : $x = 3 + 2r$, $y = 2 + r$, $z = -2 + 2r$; $r \in \mathbb{R}$.

66. L_1 : $x = 1 + 2t$, $y = -1 - t$, $z = 3t$; $t \in \mathbb{R}$.

L_2 : $x = 2 - s$, $y = 3s$, $z = 1 + s$; $s \in \mathbb{R}$.

L_3 : $x = 5 + 2r$, $y = 1 - r$, $z = 8 + 3r$; $r \in \mathbb{R}$.